

Report

1. Introduction

Dataset Description.

We work with a loan-approval dataset containing borrower demographics (e.g. age, gender, education), loan attributes (amount, interest rate, home ownership), and a binary target `loan_status` (approved vs. default).

Problem Statement.

Predict whether a loan will be paid back (`loan_status`) given borrower and loan features using K-Nearest Neighbors (KNN).

2. Data Preprocessing

1. Initial Inspection & Cleaning

- a. Loaded via `pd.read_csv()`.
- b. Checked for duplicates (`loan_data.duplicated().sum()`).
- c. Checked for null values (`loan_data.info()`)
- d. Examined distributions (`loan_data.describe()`).

2. Feature Selection

- a. Drop the label that we predict and other features that aren't so important to make X features:

```
["loan_status", "person_gender", "person_education", "person_em  
p_exp", "cb_person_cred_hist_length", "person_age", "loan_inten  
t"].
```

3. Scaling Numerics

- a. Applied `MinMaxScaler` to all `int64` and `float64` features for KNN distance consistency.

4. Encoding Categoricals

- a. One-Hot Encoded

```
["person_home_ownership", "previous_loan_defaults_on_file"]
```


via a `ColumnTransformer`.

5. Train/Val/Test Split

- a. 60 % train, 20 % validation, 20 % test using two successive `train_test_split` calls with `test_size=0.2` and then `0.25` on the remainder.

3. KNN Model

3.1 Manual Hyperparameter Search

- Loop $K = 1 \dots 20$, train on train set, evaluate on validation set:
 - Report accuracy, precision, recall, F1 for each K .
 - This gives a first sense of a good range for K .

4. Cross-Validation

- Performed 5-fold CV on the **training set only**:

Reported:

- Individual fold accuracies
- Average CV accuracy

5. Confusion Matrix Analysis

1. Confusion Matrix

- a. Displayed via `ConfusionMatrixDisplay.from_predictions` for the two models (validation and cross validation)